Source code:

```
import pandas as pd
import numpy as np
from sklearn.cluster import KMeans, AgglomerativeClustering, DBSCAN
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import silhouette_score

# Creates the dataset
data = {
    'Customer ID': ['C1', 'C2', 'C3', 'C4', 'C5', 'C6', 'C7', 'C8', 'C9',
'C10'],
    'Age': [22, 25, 45, 52, 23, 48, 35, 37, 29, 50],
    'Annual_Spending': [200, 220, 800, 850, 250, 780, 400, 420, 300, 900],
    'Purchases_Month': [5, 6, 2, 1, 7, 2, 4, 3, 5, 1]
}
df = pd.DataFrame(data)
X = df[['Age', 'Annual_Spending', 'Purchases_Month']]

# Normalization
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# K-Means
# Centroids are placed, points assigned to nearest, centroids are updated.
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
df['KMeans Cluster'] = kmeans.fit_predict(X_scaled)

# Hierarchical clustering
# Groups data in a tree-like structure
hierarch = AgglomerativeClustering(n_clusters=3)
df['Hierarchical Cluster'] = hierarch.fit_predict(X_scaled)

# Apply DBSCAN
# Groups are based on density
# eps=1.2 is used to account for the small sample size
dbscan = DBSCAN(eps=1.2, min_samples=2)
df['DBSCAN Cluster'] = dbscan.fit_predict(X_scaled)
```

```
print(df[['Customer ID', 'KMeans Cluster', 'Hierarchical Cluster', 'DBSCAN
Cluster']])

# Calculates the silhouette scores
km_sil = silhouette_score(X_scaled, df['KMeans Cluster'])
hi_sil = silhouette_score(X_scaled, df['Hierarchical Cluster'])

# DBSCAN check because the silhouette needs at least 2 clusters and no
noise-only
if len(set(df['DBSCAN Cluster'])) > 1:
    db_sil = silhouette_score(X_scaled, df['DBSCAN Cluster'])
else:
    db_sil = 0

# Displays the scores
print(f"K-Means Silhouette Score: {km_sil:.3f}")
print(f"Hierarchical Silhouette Score: {hi_sil:.3f}")
print(f"DBSCAN Silhouette Score: {db_sil:.3f}")
```

Output:

```
  Customer ID  KMeans Cluster  Hierarchical Cluster  DBSCAN Cluster
0          C1               0                     0               0
1          C2               0                     0               0
2          C3               1                     1               1
3          C4               1                     1               1
4          C5               0                     0               0
5          C6               1                     1               1
6          C7               2                     2               0
7          C8               2                     2               0
8          C9               0                     0               0
9         C10               1                     1               1
K-Means Silhouette Score: 0.624
Hierarchical Silhouette Score: 0.624
DBSCAN Silhouette Score: 0.673
```

1. Clustering Principles and Classification

a. Clustering and its Objective in This Scenario

Clustering is an unsupervised machine learning (ML) technique that is used to group similar unlabeled data points based on defined similarity measures (Google for Developers, 2025). In this scenario, the primary objective is market segmentation. Natural groupings of customers based on specific features, such as age, spending, and purchase frequency, are used to curate marketing strategies. The model will identify the relationship between features to predict the categories of new customers.

b. Classification and How it Works

In contrast to clustering, classification is a supervised approach where data is already labeled with their categories (Murel & Kavlakoglu, n.d.). If the store had historical data where customers were already labeled, a classification model could be trained on that. It would learn the relationships between features and labels to predict the categories of new customers. However, classification does not apply to this scenario because the data is unlabeled, meaning it lacks predefined customer categories.

c. Clustering vs. Classification

The primary difference between clustering and classification is the presence of labels. Clustering finds patterns in unlabeled data, while classification maps inputs to predetermined labels (Murel & Kavlakoglu, n.d.). Clustering is more suitable for this dataset because the online retail store does not have predefined customer categories. The store can create meaningful categories associated with customer behavior.

2. Applying Clustering to the Dataset

a. Data Preparation

To prepare the dataset, “Customer ID” should be dropped because it is a non-numeric identifier. The other features (Age, Annual Spending, Purchases per Month) should be normalized or scaled to ensure that features with larger numerical ranges do not assert a disproportionate influence on distance calculations (W3Schools, n.d.). For the value of k, a choice of k = 3 is suitable to capture low, medium, and high engagement for the small dataset.

b. Key Steps Involved in Forming Clusters

For the application of the K-means clustering algorithm, refer to the source code and output. As for the steps involved in forming clusters, the first is initialization, which is where k centroids are randomly selected (e.g., k =3) (Google for Developers, n.d.). The next step is assignment, which, in this scenario, assigns each customer to the nearest centroid to form initial clusters. After that, the update step is where the centroid for each cluster is recalculated by taking the mean position of all assigned points (Google for Developers, n.d.). Finally, convergence focuses on repeating the assignment and update steps until cluster membership stops changing.

c. Implementation of Hierarchical Clustering and DBSCAN

Refer to the source code and output regarding the implementation of hierarchical clustering and DBSCAN. Comparing the K-means results to hierarchical clustering and DBSCAN reveals some interesting observations. K-means generally achieves a high silhouette score, which is often greater than 0.6. This shows that the segments are very distinct. K-means

works best when clusters are relatively well-separated. Hierarchical clustering builds a tree-like structure called a dendrogram by merging the closest pairs of points until one large cluster remains, unlike K-means, which assumes clusters are roughly circular and compact (Tutorialspoint, 2024). DBSCAN differs in the fact that it is density-based, where closely packed points are grouped together, and points in low-density areas are identified as noise or outliers (Dey, 2026).

3. Evaluation of Clustering Performance

a. Evaluation Using WCSS, Silhouette Score, and Davies-Bouldin Index

WCSS measures how tight clusters are. The lower the value, the more compact the cluster (Reddy, 2025). Silhouette score measures how similar a point is to its own cluster compared to others, ranging from -1 to 1. The Davies-Bouldin Index (DBI) measures the average similarity between each cluster and its closest neighbor, with a lower score indicating better separation (Reddy, 2025).

Out of these metrics, the silhouette score is the most appropriate because it provides a clear and normalized validation of both cohesion and separation without the need for external labels. It will work well for determining the distinction between customer segments and their overlap.

b. Silhouette Score for the Clusters Obtained Using K-Means, Hierarchical Clustering, and DBSCAN

For the computations, refer to the source code and output.

c. Silhouette Score Comparison

Looking at the scores, K-means tends to achieve the highest silhouette score. This means that the segments are very distinct. As mentioned before, K-means works best with well-separated clusters, which matches this dataset's divide between age and spending. Hierarchical clustering performs similarly; however, K-means offers better scalability and produces more meaningful distinctions for the online retailer to act upon.

4. Reinforcement Learning

a. Key Components of Reinforcement Learning

The key components of a reinforcement learning (RL) system consist of the agent, environment, actions, and rewards. In this scenario, the agent is the smart discount system that makes decisions. The environment is the retail platform that the customers will interact with. The actions are the specific discount amounts (e.g., 10%, 15%). Finally, the rewards are positive outcomes from the environment (e.g., completed purchase).

b. How the System Learns Discount Strategy Through Interaction and Feedback

The system learns how to offer optimal discounts through interaction and feedback. It must balance the exploration-exploitation trade-off (Murel & Kavlakoglu, n.d.). In this scenario, exploration is the trialing of different discount strategies on customers, and exploitation is using strategies that have already proven successful. The agent will update its policy over time by mapping customer states to the optimal action to maximize the cumulative reward.

c. RL vs. Supervised and Unsupervised Learning

In this scenario, supervised learning would require the store to already know the correct discount for every scenario to train the model. Unsupervised learning would only group similar customers but not decide on which course of action to take. Reinforcement learning stands out in this regard because it learns through action. It does not rely on a ground truth; instead, it learns the optimal strategy by interacting with the environment directly and receiving feedback (Murel & Kavlakoglu, n.d.).

Word count: 886

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